

Improvement of Space Charge Suppression of Polypropylene for Potential Application in HVDC Cables

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ABSTRACT

High-performance polypropylene (PP) plays an important role in electrical/electronic engineering fields. Especially in high voltage direct current (HVDC) cables it compares well to cross linked polyethylene (XLPE) for its potential application as an eco-friendly material without cross-linking. The space charge injection under high electric stress is main obstacle for the development of HVDC cables. Here we adopted chemical modification on PP with polar functional group to improve the electrical properties, whereby the molecular structure has been designed to obtain excellent insulating material. The space charge suppression, dielectric properties and crystal characteristics of PP with and without grafting with maleic anhydride (MAH) were investigated. Results demonstrated that the MAH was successfully grafted onto PP macromolecular chain. Compared to pure PP, the grafting with 2 wt% MAH can effectively suppress space charge injection and provide better stability in volume resistivity as temperature increases. Besides, their dielectric properties were studied, and the mechanism of space charge suppression was proposed. This provides a useful method to prepare the HVDC cable insulating materials.

Index Terms — Space charge, trap, polypropylene, dielectrics, HVDC cables.

1 INTRODUCTION

HIGH voltage direct current (HVDC) cable plays an important role for the electric power transmission [1-3]. Compared with the oil-filled cables and mass impregnated cables commonly used as HVDC cables, the polymer insulating cables have certainly attractive characteristics such as smaller diameter, light weight, good flexibility, and low cost. However, one of the main obstacles for the application of polymer insulating HVDC cable is the space charge accumulation under high electric stress, which can cause a distortion of electric field, a generation of electrical trees, and final electrical breakdown [4-6]. Therefore, the suppression of space charge accumulation under high electric field has become one key issue in HVDC cable development [7]. To overcome the space charge injection and accumulation in HVDC polymer cables, the semi-conductive or insulating nanoparticles have been widely doped into polyolefin, such as MgO, ZnO, Al₂O₃ [8-10]. In this way, the space charge accumulation can alleviate under the electric stress through doping a small amount of nanoparticles. However, due to

huge surface energy, their agglomeration in polymer matrix seriously restrict the development of high-performance HVDC cables [11]. Furthermore, it is difficult to well modify the surface to improve the compatibility with the polymer matrix [12]. Compared to the way of doping, it can well improve the electrical properties of polyolefin by designing molecular structure which can bring great advantages to prepare the HVDC cables.

Recently, more efforts have been devoted to the development of new polymeric materials used as high performance HVDC cables [13]. It has generally been agreed on the importance of suppressing space charge accumulation. Among the candidates, compared to cross-linked polyethylene (XLPE), polypropylene (PP) has higher operation temperature range and good recycling properties which involves that PP has high potential application as eco-friendly advanced DC cable material [14, 15]. Therefore, a deep understanding of the relation between space charge and chemical modification in PP should be investigated, which is still an important topic.

Two kinds of traps usually exist in the polymer insulating materials, namely, the deep traps and shallow traps [16]. On one hand, deep traps can capture carriers (e.g. electrons and/or holes) to form an independent electrical field. Depending on the

polarity of this independent charge-induced field vs. the polarity of the applied (external) field, such independent field could decrease the effective electrical field which would then weaken the real electric field applied on the materials. [17-19]. The trap affected by chemical modification should be studied to investigate the mechanism of space charge suppression.

To develop the HVDC cables, we should also pay an attention to the electric field distribution caused by the variation of resistivity under the range of operation temperature [20]. The polymeric cable is a poor conductor of heat, so there is the temperature gradient that from inside cable to outside. It is very important for us to study the variation of resistivity under different temperatures.

Here we adopted the method of chemical modification on PP to improve its property of space charge suppression under high electrical stress, which is different from the method of doped nanoparticles. The Maleic anhydride (MAH), with carbonyl polar functional group, was grafted onto the PP chain. The influence of functional group of carbonyl on the motion of molecular chain and trap formation were studied, which could effectively suppress the space charge injection and accumulation. Besides, the stability of resistivity with the increase of the temperature was also discussed. Furthermore, the dielectric properties were investigated to evaluate the property of PP-g-MAH. It is believed that this work could provide a new strategy to prepare an eco-friendly advanced material for HVDC cables.

2 EXPERIMENT PROCEDURES

2.1 MATERIALS AND SAMPLES PREPARATION

Polypropylene pellets with a melt flow rate of 3 g/10 min, $M_w \approx 400,000$ and $M_n \approx 50,000$ (Yanshan Petrochemical Co., China) were used in this study. Maleic anhydride (MAH), KOH, xylene, thymol, HCl and dicumylperoxide (DCP) were provided by Fushun 3rd Chemical Factory.

The PP-g-MAH was prepared by melt grafting route and followed by compressive molding technology using flat vulcanizing machine [21]. Firstly, PP pellets, MAH and DCP were dry blended in the stirred tank. Then the mixtures were extruded using twin-screw extruder ($\Phi=21.7\text{mm}$, $L/D=42$) with the screw speed of 30 rpm at 190 °C. In order to remove the ungrafted MAH and impurities, the extruded mixtures were dissolved in refluxing xylene at 120 °C and washed in acetone. Then, the PP grafted with MAH (PP-g-MAH) was obtained.

Subsequently, the PP-g-MAH was placed in a stainless steel mold and compressively molded using flat vulcanizing machine with a pressure of 40 MPa at 190 °C for 15 min. Finally, the PP-g-MAH in the thickness of $\sim 225\text{ }\mu\text{m}$ was prepared. The PP, PP-g-1 wt% MAH and PP-g-2 wt% MAH samples for all the tests are at the same preparation procedure and conditions.

2.2 CHARACTERIZATION AND TESTS

Crystal structures of PP, PP-g-1 wt% MAH and PP-g-2 wt% MAH were characterized by an X-ray diffractometer

(XRD) using Cu-K α radiation (Rigaku, D/MAX-RB) and the crystal morphology observation by polarization microscope (OLYMPUSBX 51). Chemical structures of the samples were characterized by Fourier transform infrared spectroscopy (FT-IR, Nicolet 5700). Thermal characteristic was tested by differential scanning calorimetry (DSC) after eliminating the thermal history with the velocity of 5 °C /min. Space charge distribution of the samples was measured using pulsed electro-acoustic (PEA) method. The volume resistivity and the current density were investigated by high resistivity instrument (KEITHLEY 6517B). Trap level of the samples was measured by thermally stimulated current (TSDC) method. Prior to measurement, the samples with the thickness of 225 μm were evaporated with gold and polarized with a DC electric field of 40 MV/m for a polarization time of 20 min in dry nitrogen at room temperature and then quickly cooled down to -80 °C, afterwards electric field is removed at heating rate of 3 °C /min from -80 to 70 °C. Finally, the trapped space charges were obtained by current measurement across the samples. Dielectric properties of PP and PP-g-MAH films were measured by a precision impedance analyzer (Agilent 4294A) in the frequency range from 10^3 Hz to 10^6 Hz .

The amount of MAH grafting was done by titration method by Gaylord [22]. The 0.3 g of PP-g-MAH was dissolved in 150 mL water-saturated xylene at 108 °C, and then titrated with 0.05 N ethanolic KOH using thymol blue indicator. Finally, a 0.5–1.0 mL excess of KOH solution was added and the deep blue color was back-titrated to a yellow end point by the addition of 0.05 N isopropanolic HCl to the hot solution. Finally, we got the grafting rate of MAH of PP-g-1 wt% and PP-g-2 wt%.

3 EXPERIMENT RESULTS

3.1 THE PP GRAFTED WITH MALEIC ANHYDRIDE

Figure 1 shows the FT-IR spectrum of PP grafted with maleic anhydride. A spectrum of PP (without grafting) is also shown in the figure as a comparison. As can be seen in Figure 1, the absorption peaks at 1455 cm^{-1} and 1375 cm^{-1} are attributed to the C-H bending vibration of PP. The characteristic absorption peak at 1782 cm^{-1} is attributed to the carbonyl group in maleic anhydride, revealing its successfully grafting onto PP.

Figure 2 illustrates the effect of maleic anhydride on melting temperature of PP. The melting temperature of PP was about $\sim 168\text{ }^\circ\text{C}$, while the melting peak is at about $164\text{ }^\circ\text{C}$ as shown in Figure 2. However, with increasing the content of MAH grafting onto PP, the melting peak was decreased. It means that the thermal properties of PP-g-MAH are greatly dependent on the content of grafting MAH. Only one melt temperature peak in PP-g-MAH can be seen, showing that the maleic anhydride were successfully graft onto PP.

As shown in Figure 2, the pure PP at $145\text{ }^\circ\text{C}$ has a melt peak, this small peak might be attributed to the β -crystal form, and from the XRD results, we can also observe the β -crystal, as shown in Figure 3. Furthermore, the peak was disappeared

after grafting MAH onto the PP, suggesting that the MAH can affects the rearrangement of molecular chain, which contributes to the crystal formation.

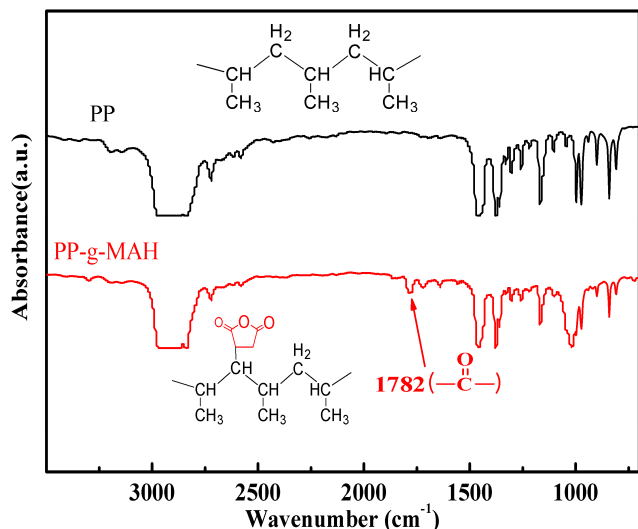


Figure 1. FT-IR spectrum of PP and PP-g-MAH.

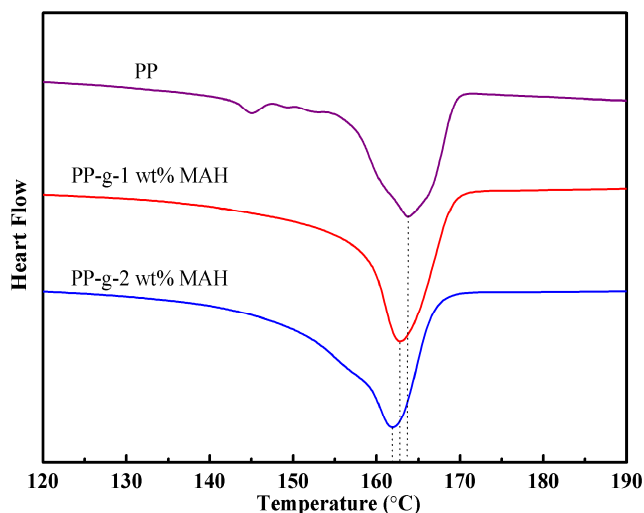


Figure 2. DSC spectrum of PP and PP-g-MAH.

The XRD patterns of PP and PP-g-MAH are shown in Figure 3. There are obvious differences for the in crystal forms between PP and PP-g-MAH. From curve 1 in Figure 3, both the α - and β -crystal forms are observed distinctively. The characteristic diffraction peaks at $2\theta=14.2^\circ$, 17.1° , 18.6° , 21.2° and 21.9° , which respectively correspond to the α (110), α (040), α (130), α (111) and α (131) crystal planes of α PP. The characteristic diffraction peaks at $2\theta=16.2^\circ$ which corresponds to the β (300) plane, suggesting the presence of β -crystal in PP. However, for the PP-g-MAH (Figure 3, curve 2), there is only α -crystal form from the position of diffraction peaks at $2\theta=14.2^\circ$, 17.1° , 18.6° , 21.2° and 21.9° . It may be due to the maleic anhydride successfully grafted onto PP, which affects the motion of molecular chain and the formation of crystal.

In this work, PP and PP-g-MAH samples for all the tests were prepared under the same conditions in order to eliminate the effect of other factors on the crystal form. With the maleic anhydride graft on the molecular chain of PP, the highly polar carbonyl groups in maleic anhydride would affect the arrangement of molecular that might influence the crystal form formation. The α -crystal form is more stable than β -crystal structure. The stability of crystal form plays an important role on the insulating properties of materials.

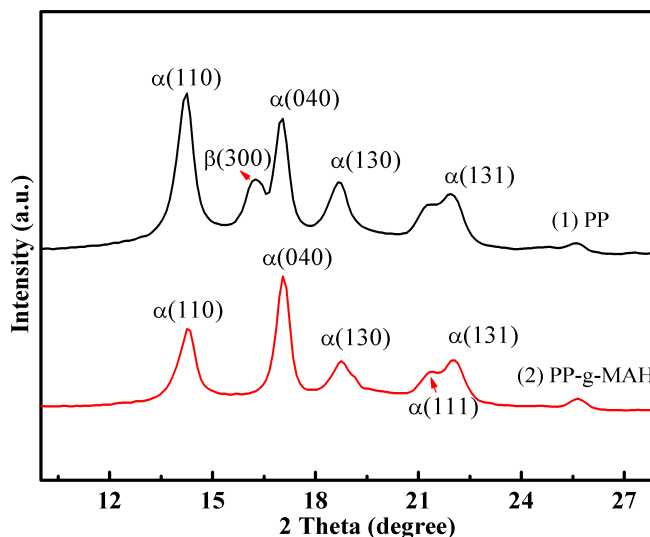


Figure 3. X-ray diffraction patterns for PP and PP-g-MAH.

The different sizes of spherulites are observed in both PP and PP-g-MAH as shown in Figure 4. The spherulites in PP are bigger than that in PP-g-MAH. This may due to the arrangement of molecular chains after grafting maleic anhydride with carbonyl group onto PP. Moreover, the interfaces between spherulites are clear as shown in sample of PP. The interfaces are easier to induce defects, which may accumulate space charges under high electrical field [23]. As can be seen in PP-g-MAH, the spherulites become smaller compared to PP.

3.2 SPACE CHARGE CHARACTERISTICS OF PP AND PP-G-MAH

Figure 5 shows the space charge distribution in the samples at high electric stresses of 65 kV/mm applied for different durations (from 10 s to 6 hours). It can be seen from Figure 5a that a few amounts of homocharges accumulate in the vicinity of the cathode and anode when the electric stress was applied for 10 s.

However, the carriers injection from cathode and anode increased with the increase of time, resulting that the space charge packet slowly grew up and accumulated. Especially, the space charge accumulation occurred from the surface of material close to electrode to the interior of material. It means that space charge packet formation move to inner material with the time increased under high electric stress, and result in the local effective electric field formation, becomes easy to breakdown.

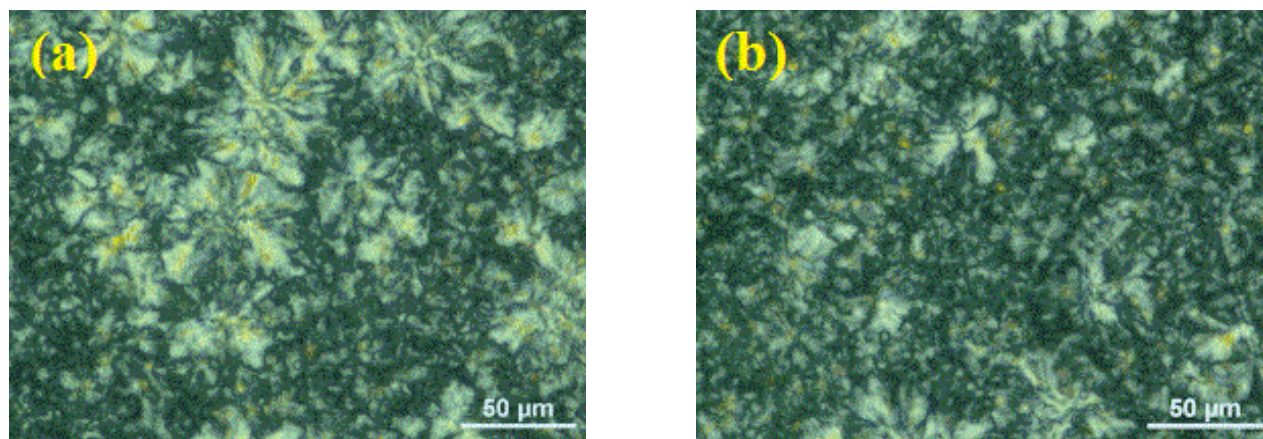


Figure 4. Polarization microscope photos at room temperature (a) PP, (b) PP-g-2wt% MAH with the thickness of 30 μm .

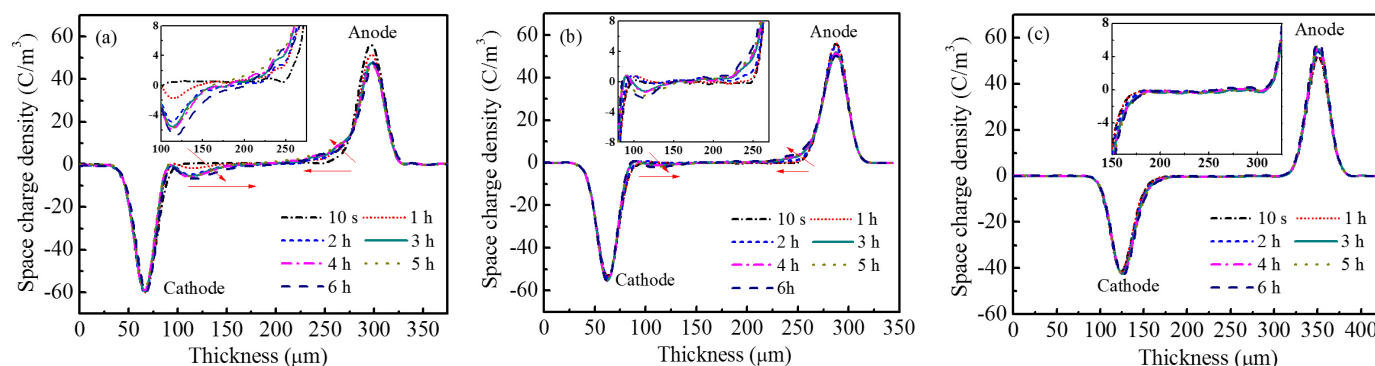


Figure 5. Space charge distribution in the samples under the applied electric field of 65 kV/mm: (a) PP, (b) PP-g-1 wt% MAH, (c) PP-g-2 wt% MAH, respectively.

Obviously, from Fig. 5b, once grafting 1 wt% MAH onto PP can slightly reduce the space charge injection from both cathode and anode, but also cannot completely suppress space charge injection. To be precisely, a small quantity of carriers injected and slowly increased with time nearby both anode and cathode, and no space charge packet can be seen. Nevertheless, the space charge injection can be hardly observed in PP-g-2 wt% MAH as shown in Fig. 6c. It means that the space charge injection has been effectively suppressed.

There are two possible reasons. (i) the functional group of MAH induces the smaller spherulites compared with the PP; (ii) the β -crystal form disappeared due to the rearrangement of molecular chain. To be more specific, the MAH grafted onto PP changed the configuration of the molecular chain, affecting the arrangement of the chain of PP. When the chain of PP-g-MAH was arranged, the new crystals would be formed so that large of traps will be induced at the crystal boundary and amorphous region [24, 25].

With increasing the content of grafted MAH on molecular chain of PP, more of crystalline and amorphous region are effected by the functional group. On one hand, the functional group of MAH existing in crystalline region will change the energy gap of crystal, and the space charge become more difficult to accumulate as shown in Figure 5c. On the other hand, the functional group of MAH exists in amorphous region so that the motion of molecular chain will be much

easier because the MAH provide more interspace compared with the pure PP, resulting in the decrease of number of traps in amorphous region. However, the violent friction of molecular chain in boundary between crystalline and amorphous region induces number of traps. The migration of carriers effected by the distribution of traps, resulting in insulating properties of amorphous region become reliable. In this case, the space charge suppression occurs in PP-g-MAH, which has great prospect application as insulating materials in HVDC cable.

The space charge distribution of the samples short-circuited at grounded condition is shown in Figure 6. As shown in Figure 6a, most space charge injected in PP belongs to deep trapping state. There is still a large quantity of space charge exiting in PP after decaying 60 s, while the trapped space charge is stable. In Figure 6b, the trapped space charge dissipates a little as the time increased and large amounts of space charge are in trapped states. However, Figure 6c shows that little space charge exited in PP-g-2 wt% MAH, indicating that MAH grafting onto PP can suppress space charge injection and nearly a little space charge observed on the interior of PP-g-2 wt% MAH.

3.3 THE TRAP LEVEL DENSITY AFFECTED BY GRAFTING MAH ONTO PP

To reveal the relationship between variation of molecular chain and trap distribution affected by the grafted maleic

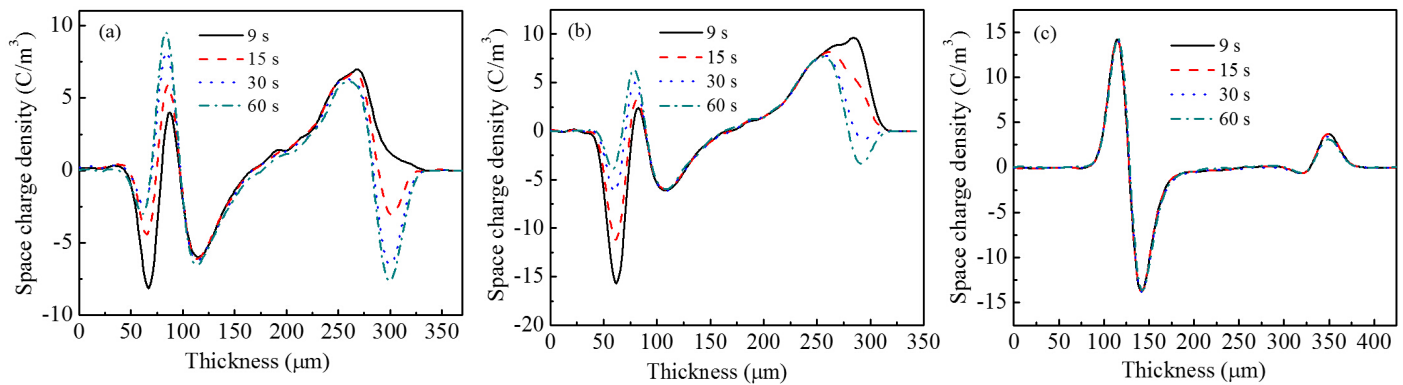


Figure 6. Space charge distribution in samples at the shorted-circuit condition: (a) PP, (b) PP-g-1 wt% MAH, (c) PP-g-2 wt% MAH, respectively.

anhydride onto PP, the thermally stimulated discharge current (TSDC) measurement was adopted to investigate the trap level density variation [26, 27]. Precisely, the motion of molecular chain influenced by grafted functional group onto the chain of PP, which would deeply affect the formation of crystal and trap. Most of traps mainly exist in crystal boundary, due to the violent friction of molecular chain in boundary. In order to observe the trap level distribution, the discharge current with temperature change were transformed into trap level density and trap level, respectively, as shown in Figure 7 [28]. Meanwhile, the thermally stimulated current spectrum was inset into Figure 7. The discharge current were obtained from -60 to 70 °C at a heating ratio of 3 °C/min.

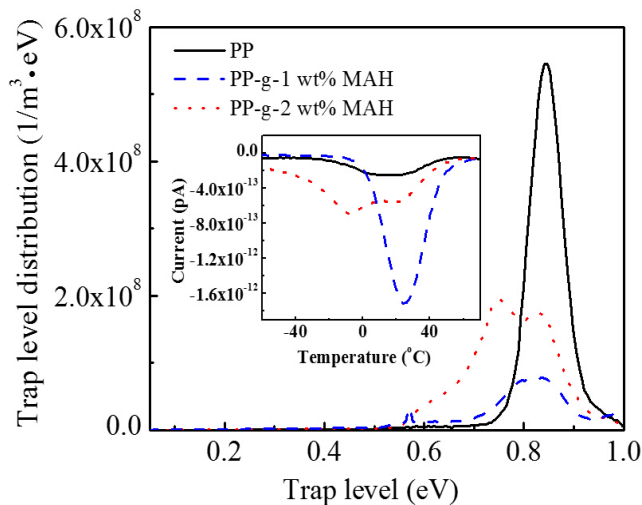


Figure 7. Trap level density in PP and PP-g-MAH, the inset is thermally stimulated current spectrum.

It can be seen from Figure 7 that grafting MAH onto the PP did not induced new trap level and the trap level was almost in the range from 0.65 to 0.9 eV. This may be due to that the functional group of MAH grafted onto PP as a whole chain to rearrange and form the crystalline and amorphous again. However, after grafting 1 wt% and 2 wt% MAH onto chain of PP, the trap level density was increased. Compared to pure PP, the trap level density of PP-g-1 wt% MAH and PP-g-2 wt% MAH were increased one times and 5 times, respectively. This phenomenon may be contributed to the

motion of molecular chain between crystalline and amorphous region, resulting in the numbers of traps in boundary.

3.4 THE VARIATION OF VOLUME RESISTIVITY UNDER DIFFERENT TEMPERATURE

The volume resistivity of insulating materials is very important for designing the DC cables. The electric field distribution is relates with resistivity for DC cable, which is different from the alternating current (AC) cables that the electric field distribution is determined by permittivity. Moreover, the volume resistivity is deeply influenced by temperature, and the motion of molecular chain is also affected by the temperature distribution.

Figure 8 shows the volume resistivity of samples decreased with the increased temperature. It means that the conductivity would increase with temperature increased. However, the descent rate of the samples with different content of MAH was not remarkable. To be specific, the pure PP is violently affected by temperature, the grafting MAH onto PP can improve the property of heat-resisting. Evidently, the PP-g-2 wt% MAH has better stability of volume resistivity attribute to the MAH limited the electronic hopping.

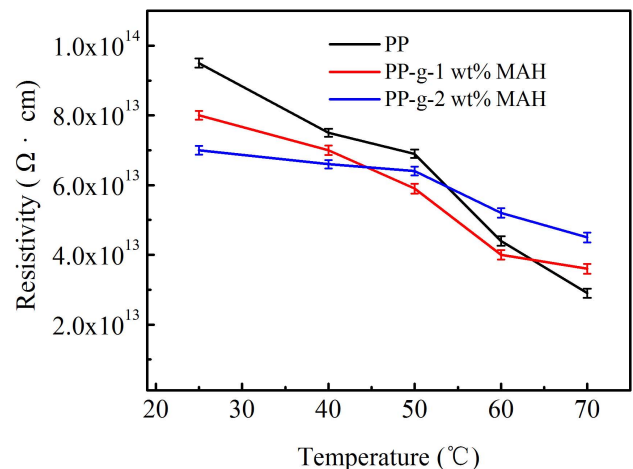


Figure 8. Temperature dependence of volume resistivity of PP with and without grafted MAH.

3.5 THE DIELECTRIC CHARACTERISTICS OF PP AND PP-G-MAH

Figure 9 shows the dielectric permittivity and losses of PP and PP-g-MAH in the frequency range of 10^3 to 10^6 Hz at room temperature.

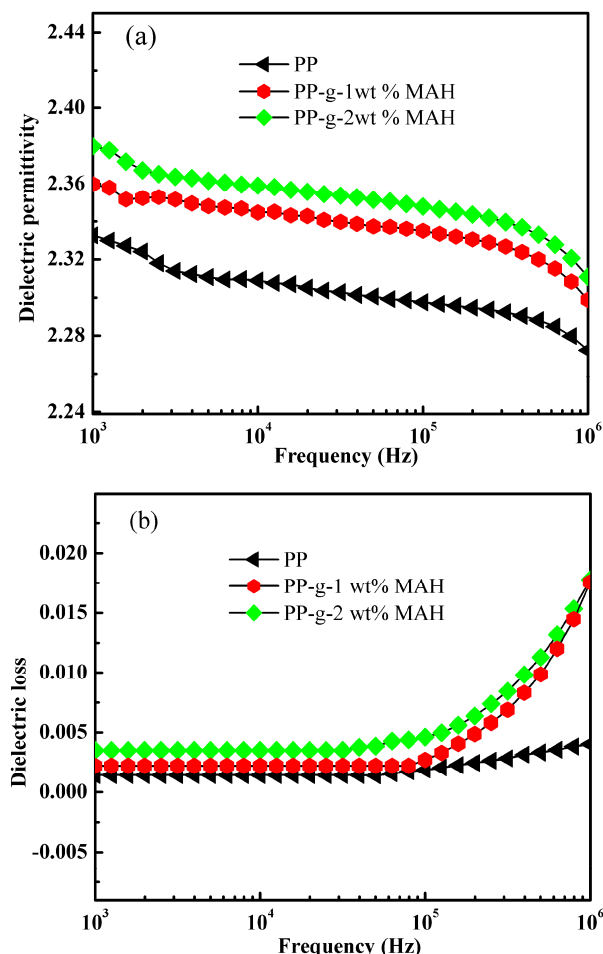


Figure 9. Frequency dependence of (a) dielectric permittivity and (b) dielectric losses of PP with and without grafted MAH.

It can be seen from Figure 9a, the dielectric permittivity of PP is about 2.3, while it was only increased to 2.34 and 2.37 with 1 wt% and 2 wt% of MAH grafted onto PP, respectively. The MAH can not effectively improve the dielectric permittivity of PP. Meanwhile, the dielectric permittivity of the samples decreases with the increasing frequency which is due to that the motion of polar groups could not catch up the change of frequency of electric field at high frequency range. Figure 9b shows the dielectric losses of the samples vs frequency. In the range of 10^3 to 10^5 Hz, the dielectric losses of pure PP is lower than that of PP-g-MAH which also keep low (<0.004). A small increase of dielectric losses of all the samples can be observed when the frequency is up to 10^6 Hz. This may be due to the increase of relaxation times which bring the energy loss. Therefore, the dielectric losses peak moves towards the high frequency, and the low dielectric loss of the PP-g-2 wt% MAH can be satisfied with the application of DC cables.

4 CONCLUSIONS

In summary, by grafting with maleic anhydride onto the molecular chain of PP, the functional group of MAH induces smaller spherulites compared to pure PP, the β -crystal form disappeared due to the rearrangement of molecular chain. Meanwhile, with increasing the content of grafting MAH, the performance of suppressing space charge injection and accumulation in PP-g-MAH is much better than that in PP under the 65 kV/mm. Moreover, the stability of volume resistivity of PP-g-MAH with increased temperature also reveals an enhanced improvement, compare to the pure PP. The dielectric permittivities of PP-g-1 wt% MAH and PP-g-2 wt% MAH did not largely increase and the dielectric losses also keeps low, which can satisfy the materials of both low permittivity and low dielectric losses. This work will promote the development of insulating materials for HVDC cables application.

ACKNOWLEDGMENTS

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